AESC v2.3: Two-Speed Memory Architecture

(Revised Technical Report — Memory Fixes, Brain-Inspired Metacognition, Contribution Results)

Oct 9, 2026 · Giridhari Karmakar · giriisdev@gmail.com

Code: https://github.com/giridhari-code/AERIS_64 (commit 70d54bd)

Abstract

AESC (Adaptive Event-State Computation), also released as NeuroField, is a recurrent language model with a k-step neural field, a predictive dendrite, top-k skill routing, and two-speed memory. Fast weights are written token by token by a surprise-gated delta rule; slow weights are trained by backpropagation and surprise-weighted replay.

The previous revision (v2 / Oct 2, 2026) fixed eleven failure modes from the first 10M implementation and reported that a 9.92M reference model solved 4-pair associative recall at 1.000 accuracy after 800 steps. This v2.3 report documents additional production defects that prevented the public codebase from reproducing that result, the fixes applied, measured module contributions, a brain-inspired metacognition controller, and a session-isolated tool sandbox. After the memory-path fixes, a d_model=64 production NeuroField reaches **0.973** eval accuracy on 4-pair recall at 800 steps with fast memory on, versus **0.344** with fast memory off (chance 0.125).

1. Introduction and status of claims

This report makes no claim about how a small model compares with a much larger transformer. What changed in v2.3 is that the production repository now exhibits a clear, measurable contribution from fast memory on the same synthetic recall task used in Section 7 of v2.

Expectation (unchanged): on plain next-token loss a larger transformer is expected to win; the goal is to narrow the gap.

2. Contributions of this revision

• Diagnosis and fix of production memory-path defects that nullified fast-memory contribution in the public code (Section 4).

• Measured ablation of fast memory, slow memory, neuromodulator, and metacognition on 4-pair recall (Section 5).

• Brain-inspired metacognition: monitor (surprise, conflict, confidence, arousal) and control (k, encoding bias, consolidation request) (Section 6).

• Session-isolated tool sandbox aligned with safety controls (Section 7).

• Updated design checklist and open items (Sections 8-9).

3. Architecture (unchanged data path)

Token order remains: embedding -> dendrite (local window + delayed error) -> metacognition -> k-step neural field -> predictor -> top-k skill router -> neuromodulator gate -> fast write/read -> slow read -> logits. Only the fast matrix is written by a local delta rule; slow weights consolidate via surprise-weighted replay. Timing remains leakage-safe: nothing computed from x(t+1) reaches the logits that predict x(t+1).

Fast write (Appendix A form): k_t = normalize(W_k d_{t-1}), v_t = W_v o_t, M_t = Lambda M_{t-1} + g_t k_t (v_t - M_{t-1}^T k_t)^T. Read: q_t = normalize(W_q d_t), r_t = M_t^T q_t.

4. Additional failure modes fixed in v2.3

Rows 1-11 remain as in v2. The following defects were present in the production tree after v2 and prevented fast memory from contributing on recall until fixed.

#	Failure mode	Evidence	Fix
12	Decay too fast (lambda~0.73 on some dims)	Associations wiped before query	lambda init ~ 0.97-0.998 (linspace 3.5 to 6.5)
13	W_q random vs W_k	Poor retrieval early	W_q <- W_k at init
14	Write on zero d_prev	Garbage first write	Skip write if ||d_prev||~0
15	Slow M_s xavier noise	Noise masks fast path	M_s = 0; P_s = 0 init
16	Gate starved (bias=0)	Weak writes early	bias=1.0, a_s=3.0
17	bar_S near zero	rel_S explosion	bar_floor=0.05; median k
18	No slow replay in ablation runs	Slow path unused	replay every 50 steps

Table 1. Production defects fixed in v2.3 (in addition to v2 rows 1-11).

Isolated write-read unit check after the fix: cosine similarity between stored value projection and read vector is 1.000 under a single write with g=1.

5. Contribution experiments

Task: 4 key-value pairs, 8 keys and 8 values, answer at the last position (chance 0.125). Model: production NeuroField, d_model=64, d_k=d_v=32, 4 skills, top-k=2, AdamW lr=1e-3, batch 8, CPU, single seed unless noted. Ablations use AblationConfig switches.

5.1 Before vs after memory-path fix (800 steps)

Condition	Eval acc (before)	Eval acc (after)
full (all modules)	0.383	0.973
no_fast_mem	0.355	0.344
Delta (full - no_fast)	+0.028 (null)	+0.629 (clear)

Table 2. Fast-memory contribution appears only after v2.3 fixes. Train best reaches 1.000 on full.

5.2 Module ablations at 400 steps (after fix)

Ablation	Eval acc	Delta vs full
full	0.555	-
no_fast_mem	0.426	-0.129
no_slow_mem	0.543	-0.012
no_neuromod	0.438	-0.117
no_metacog	0.367	-0.188
skills_only	0.395	-0.160

Table 3. Fast memory, neuromodulator, and metacognition help on pure recall; slow memory is near zero (expected). Single seed — multi-seed GPU runs required before strong claims.

These results support the mechanistic claim that the fast delta-rule path is necessary for high recall accuracy in this implementation. They do not include a comparison against transformer baselines.

6. Brain-inspired metacognition

v2 metacognition was a minimal rule set: relative surprise baseline and a discrete k schedule. v2.3 expands it into an explicit monitor-control loop without claiming biological fidelity.

Monitor. Relative surprise S_rel = S / bar_S. Conflict from previous-step router entropy. Confidence = sigmoid(-(S_rel - 1)). Arousal = 0.7*S_rel + 0.3*conflict.

Control. Field depth k increases with surprise and conflict. Encoding bias delta tags writes (high surprise / low confidence raises gate bias). Neuromodulator: g = sigmoid(a_s*(S_rel - 1) + a_c*|h| + b + delta). Consolidation request want_replay fires after a cumulative surprise budget and minimum token count; the trainer runs slow replay and acknowledge_replay.

The controller remains rule-based with calibration scalars. A learned metacognitive policy is future work.

7. Safety and tool sandbox

v2.3 adds a production subprocess sandbox for tool use, aligned with session isolation:

• Per-session work directory and call budget; TTL eviction of idle sessions.

• Binary allow-list; network disabled by default; CPU / address-space / wall-clock limits; audit log.

• HTTP: POST /v1/tools/shell, /v1/tools/python; GET /v1/tools/audit; POST /v1/tools/reset.

• Model session reset also drops the sandbox session.

Research-grade isolation (subprocess + rlimits), not a multi-tenant hardware jail. Red-team evaluation of memory poisoning remains future work.

8. Updated design checklist (additions)

• lambda on fast memory remains above ~0.95 on all key dims at init.

• W_q and W_k aligned at init; write skipped when d_prev is the zero state.

• Slow M_s and P_s do not inject random noise before the first replay.

• Neuromodulator gate mean at init is materially above 0.5 on unit Gaussian state.

• Metacognition exposes k, gate_bias, confidence, arousal; bar_S has a positive floor.

• Contribution ablation reports full vs no_fast_mem on 4-pair recall before other claims.

• Sandbox audit fields present when tools are enabled.

9. Limitations and non-claims

• No comparison against transformer or Mamba baselines is reported.

• Reported ablations are single-seed CPU runs on one synthetic task.

• Slow memory is not stressed by 4-pair recall; continual learning and long-copy are needed.

• Metacognition is not a learned policy and is not a claim of biological realism.

• Sandbox is not a full multi-tenant secure enclave.

• Demo checkpoints in the repo remain tiny memorisers, not assistants.

• Token-by-token recurrence is slower to train than a same-size transformer; scale GPU metrics pending.

10. Related work (pointer)

Precedents unchanged from v2: fast weights (Schmidhuber; Ba et al.; DeltaNet), surprise-driven test-time memory (Titans), state-space models (Mamba), predictive coding (Rao and Ballard), complementary learning systems (McClelland et al.), looped computation (Universal Transformers), sparse routing (Shazeer et al.). The contribution remains the combination, audit signals, and measured production fixes.

11. Conclusion

v2.3 closes a gap between the v2 reference claim and the production codebase: after correcting decay, key/query alignment, slow-path noise, gate starvation, and metacognitive baselines, fast memory shows a large, necessary contribution on associative recall (0.973 vs 0.344 at 800 steps in the reported configuration). Brain-inspired metacognition and a tool sandbox extend the control and safety surface. The next decisive experiments are multi-seed, compute-matched comparisons against transformers and Mamba on the full Section 7 task suite.

Code and configs: https://github.com/giridhari-code/AERIS_64 · Contact: giriisdev@gmail.com